Marketing Campaign Performance Analytics using Power BI

1. Introduction

Marketing teams run campaigns across several channels and brands, and they need a clear view of what those campaigns deliver in return for the money spent. Business intelligence tools help turn large volumes of raw campaign records into interactive dashboards that support quick, informed decisions. Power BI is one such tool: it connects to data, cleans and models it, and presents the results as interactive visuals.

This project analyses multi-brand marketing campaign data for three beauty and lifestyle brands (Nykaa, Purplle and Tira) using Microsoft Power BI. The dataset contains about 167 thousand campaign records covering impressions, clicks, leads, conversions, spend and revenue. The dashboard tracks key performance indicators (KPIs), compares campaign types, channels, audiences and brands, and follows the customer funnel from impressions to conversions, so that stakeholders can monitor performance and decide where to invest.

2. Objectives

- To import, clean and combine multi-brand campaign data in Power BI using Power Query
- To design a star-schema data model with a date table and a bridge table for multi-channel campaigns
- To apply DAX functions to build business measures such as ROAS, CTR, cost per conversion and growth
- To create interactive dashboards using varied charts, slicers and navigation
- To analyse revenue, efficiency and the conversion funnel, and derive insights through case studies

3. Additional Features Implemented

Beyond standard charts, the following Power BI features were added to make the report more interactive and easier to read. The reason for each is given below.

1. Drill-through

City was set as a drill-through field on the Performance and Funnel pages. A user can right-click a city on the Overview map, choose Drill through, and open the Performance or Funnel page already filtered to that city. A Back button on each page returns the user to the starting page.

Why implemented: It lets a viewer move from a high-level overview to city-level detail in one step, without re-setting slicers, and keeps the three pages connected as one analysis.

2. Page navigator buttons

A page navigator (Overview, Performance, Funnel) sits at the top of every page, and the active page is highlighted in a different colour.

Why implemented: It gives the report an app-like feel and removes the need to use the page tabs at the bottom of Power BI.

3. Synced slicers across pages

The Channel, Date, Campaign Type, City and Brand slicers are synchronised, so a selection made on one page carries over to the others.

Why implemented: A user who filters to one brand or period keeps the same view while moving between pages, which avoids inconsistent comparisons.

4. Conditional formatting

Colour rules were applied to the Avg ROI by Audience and Brand matrix (a red-to-green gradient) and to the font colour of the growth and cost cards.

Why implemented: Colour lets the viewer spot the strongest and weakest cells at a glance instead of reading every number.

5. Reference lines and targets

A dashed average line was added to the Cost per Conversion by Channel chart, and a target marker (3.00) was set on the Avg ROI and ROAS gauges.

Why implemented: A benchmark makes it clear whether a channel or metric is better or worse than average or than the goal.

6. Tooltips

Additional measures were added to the tooltips of selected charts so extra detail appears when the user hovers over a bar, bubble or slice.

Why implemented: It adds context without crowding the page with more visuals.

4. Dataset Used

Dataset: Multi-Brand Marketing Campaign Performance Dataset

Source: Kaggle – kaggle.com/datasets/sshriya08/multi-brand-marketing-campaign-performance-dataset

The data is supplied as three CSV files, one for each brand (Nykaa, Purplle and Tira). Each file has 55,555 campaign records, giving 166,665 records in total. The campaigns run from 1 July 2024 to 24 June 2025.

Dataset Features:

- Campaign ID, Campaign Type (Email, Influencer, Paid Ads, SEO, Social Media)
- Target Audience and Customer Segment
- Channel Used (single or multiple channels per campaign)
- Duration and Language
- Impressions, Clicks, Leads and Conversions
- Revenue and Acquisition Cost
- ROI and Engagement Score
- Campaign Date

Fields added during data preparation:

- CampaignKey – a unique key for each campaign record, used to link the fact table to the bridge table
- City – a derived field added during data preparation, used for the map, the City slicer and drill-through
- Date attributes (Year, Month, Month Year, Quarter) – generated in the Dim_Date table

5. Methodology (Step-by-Step in Power BI)

Step 1: Load Data

- Open Power BI Desktop
- Click **Get Data → Text/CSV**
- Select the Nykaa campaign file and load it; repeat for the Purplle and Tira files
- Click **Transform Data** to open the Power Query Editor

The three files appear as three staging queries (nykaa_campaign_data, purplle_campaign_data and tira_campaign_data), which are kept as the raw source for the later steps.

Step 2: Data Cleaning and Transformation

In **Power Query Editor**, the following steps were carried out:

1. **Appended** the three brand queries into a single table, Fact_Campaigns
2. **Changed data types:** Date for the campaign date, whole or decimal numbers for the numeric fields, and text for the descriptive fields
3. **Removed blank rows** and null values
4. **Removed duplicate** records

5. **Renamed columns** to clear, consistent names
6. **Trimmed and standardised text** values (for example brand, channel and campaign type names)
7. **Added a unique CampaignKey** column to identify each campaign record
8. **Added the City** column as a derived field
9. **Split the multi-channel values** in Channel_Used into separate rows, creating the Bridge_CampaignChannel table
10. **Created dimension tables** (Dim_Brand, Dim_CampaignType, Dim_Audience, Dim_Channel, Dim_City) by keeping only the relevant column and removing duplicates (Remove Other Columns, then Remove Duplicates)
11. Clicked **Close & Apply**

Query	Type	Purpose
nykaa / purplle / tira _campaign_data	Staging	Raw brand files kept as the source
Fact_Campaigns	Fact	Combined campaign records with all metrics
Bridge_CampaignChannel	Bridge	One row per campaign and channel, to handle multi-channel campaigns
Dim_Brand	Dimension	Unique list of brands
Dim_CampaignType	Dimension	Unique list of campaign types
Dim_Audience	Dimension	Unique list of target audiences
Dim_Channel	Dimension	Unique list of channels
Dim_City	Dimension	Unique list of cities
Measures	Measures table	Holds all DAX measures

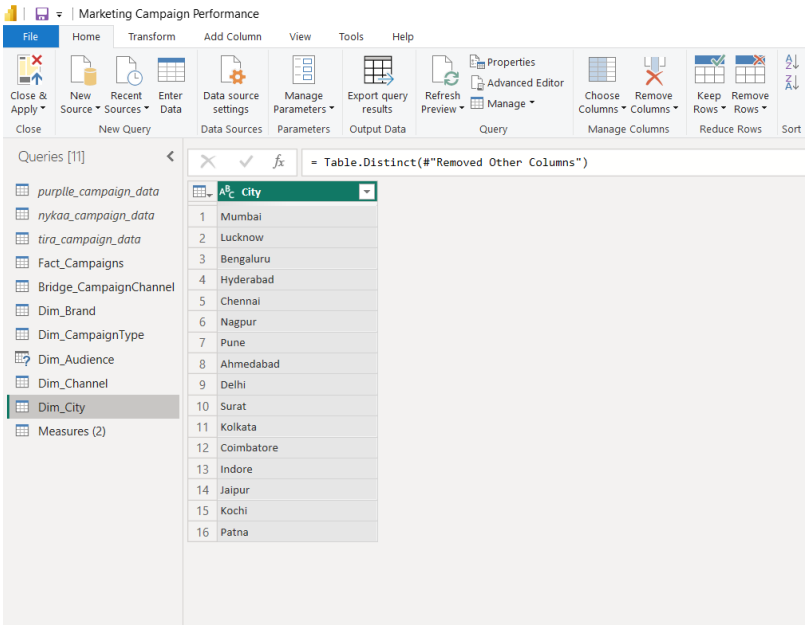


Figure 1: Power Query Editor showing the staging, fact, bridge and dimension queries

Interpretation: Separating the data into one fact table, one bridge table and five dimension tables keeps the model clean and avoids repeating text values across 166,665 rows.

Step 3: Build the Data Model

3.1 Date table (DAX). A calendar table was created from the earliest to the latest campaign date, with columns for year, month and quarter. The Year Month Number column is used to sort month names in calendar order.

```
Dim_Date =
VAR MinDate = MINX ( Fact_Campaigns, Fact_Campaigns[Date] )
VAR MaxDate = MAXX ( Fact_Campaigns, Fact_Campaigns[Date] )
RETURN
ADDCOLUMNS (
    CALENDAR ( MinDate, MaxDate ),
    "Year", YEAR ( [Date] ),
    "Month Number", MONTH ( [Date] ),
    "Month", FORMAT ( [Date], "MMM" ),
    "Month Year", FORMAT ( [Date], "MMM yyyy" ),
    "Quarter", "Q" & FORMAT ( [Date], "Q" ),
    "Year Month Number", YEAR ( [Date] ) * 100 + MONTH ( [Date] )
)
```

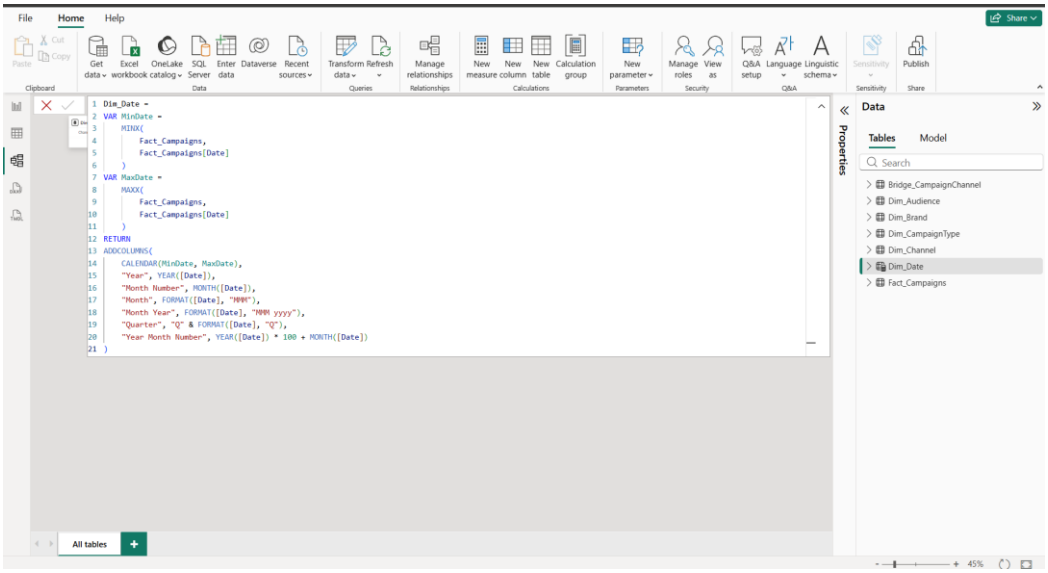


Figure 2: Creating the Dim_Date table using DAX

Interpretation: Generating the date table from the data itself means the calendar always covers exactly the campaign period and supports time-based measures such as month-over-month growth.

3.2 Relationships. The tables were connected as a star schema, with Fact_Campaigns at the centre and the dimension tables around it.

From (Fact / Bridge)	To (Dimension / Fact)	Cardinality	Filter direction	Status
Fact_Campaigns[Brand]	Dim_Brand[Brand]	Many to one	Single	Active
Fact_Campaigns[Campaign_Type]	Dim_CampaignType[Campaign_Type]	Many to one	Single	Active
Fact_Campaigns[Date]	Dim_Date[Date]	Many to one	Single	Active
Fact_Campaigns[City]	Dim_City[City]	Many to one	Single	Active
Fact_Campaigns[Target_Audience]	Dim_Audience[Audience]	Many to one	Single	Active
Fact_Campaigns[Customer_Segment]	Dim_Audience[Audience]	Many to one	Single	Inactive
Bridge_CampaignChannel[CampaignKey]	Fact_Campaigns[CampaignKey]	Many to one	Both	Active
Bridge_CampaignChannel[Channel]	Dim_Channel[Channel]	Many to one	Single	Active

Two design choices are worth noting. First, a campaign can run on several channels, so a direct link between the fact table and Dim_Channel would be many-to-many; the bridge table resolves this, and its relationship with the fact table filters in both

directions so that a channel selection reaches the campaign data. Second, the audience dimension is linked twice (Target Audience and Customer Segment), so only one relationship can be active; Target Audience is active and Customer Segment is kept inactive.

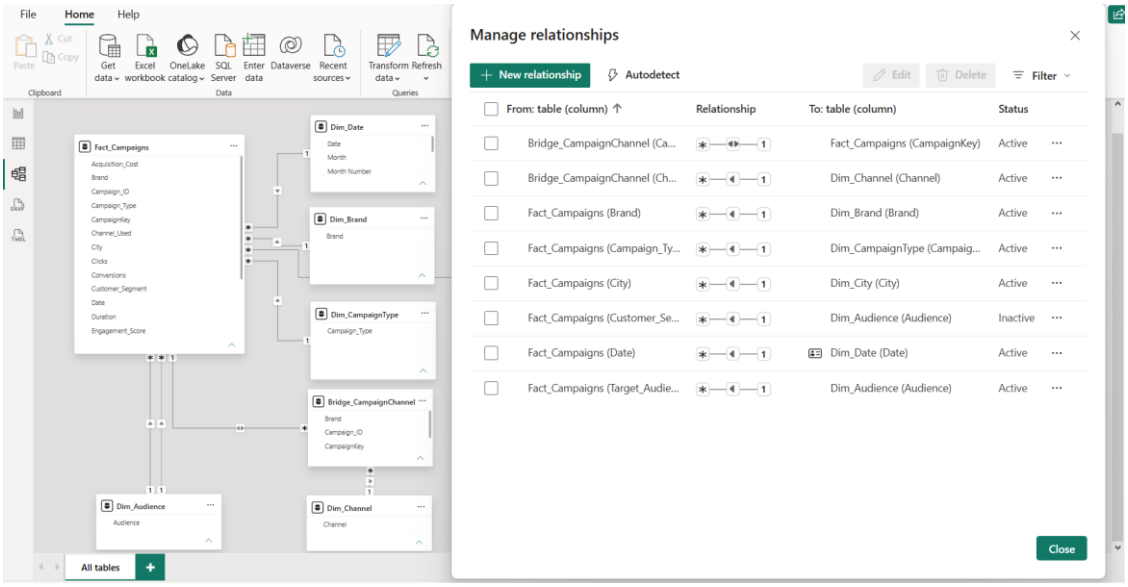


Figure 3: Manage Relationships window showing the star schema

Interpretation: The model follows a star schema, so slicers on any dimension filter the fact table correctly and all measures respond consistently.

Step 4: Create Measures (DAX)

All measures were created in a dedicated Measures table using **Model → New Measure**. Nineteen measures were created.

1. Total Revenue

Purpose: Adds up revenue for the selected filters.

```
Total Revenue =
SUM ( Fact_Campaigns[Revenue] )
```

2. Total Spend

Purpose: Spend is calculated as acquisition cost per conversion multiplied by conversions, row by row.

```
Total Spend =
SUMX (
    Fact_Campaigns,
    Fact_Campaigns[Acquisition_Cost]
        * Fact_Campaigns[Conversions]
)
```

3. Total Impressions

Purpose: Total number of times campaigns were shown.

```
Total Impressions =
SUM ( Fact_Campaigns[Impressions] )
```

4. Total Clicks

Purpose: Total clicks on campaigns.

```
Total Clicks =
SUM ( Fact_Campaigns[Clicks] )
```

5. Total Leads

Purpose: Total leads generated.

```
Total Leads = SUM ( Fact_Campaigns[Leads] )
```

6. Total Conversions

Purpose: Total conversions achieved.

```
Total Conversions =
SUM ( Fact_Campaigns[Conversions] )
```

7. Spend % of Revenue

Purpose: Share of revenue consumed by spend.

```
Spend % of Revenue =
DIVIDE ( [Total Spend], [Total Revenue] )
```

8. ROAS (Return on Ad Spend)

Purpose: Revenue earned for every rupee spent.

```
ROAS = DIVIDE ( [Total Revenue], [Total Spend] )
```

9. MoM Revenue Growth %

Purpose: Month-over-month revenue growth using variables and time intelligence (DATEADD).

```
MoM Revenue Growth % =
VAR CurrM =
    [Total Revenue]
VAR PrevM =
    CALCULATE (
        [Total Revenue],
        DATEADD ( Dim_Date[Date], -1, MONTH )
    )
RETURN
    DIVIDE ( CurrM - PrevM, PrevM )
```

10. CTR % (Click-Through Rate)

Purpose: Clicks as a share of impressions.

```
CTR % =
DIVIDE ( [Total Clicks], [Total Impressions] )
```

11. Lead Rate %

Purpose: Share of clicks that become leads.

```
Lead Rate % = DIVIDE ( [Total Leads], [Total Clicks] )
```

12. Lead to Conversion %

Purpose: Share of leads that convert.

```
Lead to Conversion % =
DIVIDE ( [Total Conversions], [Total Leads] )
```

13. Click to Conversion %

Purpose: End-to-end share of clicks that convert.

```
Click to Conversion % =
DIVIDE ( [Total Conversions], [Total Clicks] )
```

14. Conversion Rate %

Purpose: Conversions as a share of leads, calculated directly from the Leads column.

```
Conversion Rate % =
DIVIDE (
    [Total Conversions],
    SUM ( Fact_Campaigns[Leads] )
)
```

15. Cost per Conversion

Purpose: Average spend needed to win one conversion.

```
Cost per Conversion =
DIVIDE ( [Total Spend], [Total Conversions] )
```

16. Avg ROI

Purpose: Average of the ROI recorded for each campaign.

```
Avg ROI =
AVERAGE ( Fact_Campaigns[ROI] )
```

17. Avg Engagement Score

Purpose: Average engagement score across campaigns.

```
Avg Engagement Score =
AVERAGE ( Fact_Campaigns[Engagement_Score] )
```

18. Brand Revenue Share %

Purpose: A brand's revenue as a share of all brands, ignoring the brand filter (ALL).

```
Brand Revenue Share % =
DIVIDE (
    [Total Revenue],
    CALCULATE ( [Total Revenue], ALL ( Dim_Brand ) )
)
```

19. Best Campaign Type

Purpose: Returns the campaign type with the highest Avg ROI (TOPN with SELECTEDVALUE).

```
Best Campaign Type =
CALCULATE (
    SELECTEDVALUE ( Dim_CampaignType[Campaign_Type] ),
    TOPN (
        1,
        ALL ( Dim_CampaignType ),
        [Avg ROI],
        DESC
    )
)
```

DAX concepts used: aggregation (SUM, SUMX, AVERAGE), safe division (DIVIDE), variables (VAR / RETURN), time intelligence (DATEADD), context modification (CALCULATE, ALL), and ranking (TOPN, SELECTEDVALUE).

Step 5: Build Visualizations – Dashboard Creation

The report has three pages, each with four KPIs and four charts. The pages follow one story: how are we doing (Overview), what is efficient (Performance), and where do we lose people (Funnel).

How the KPI cards were created:

- Go to **Report View**
- In the Visualizations pane, select **Card (new)** or **Gauge**
- Drag the measure into the Data well (for example Total Revenue)
- Format: increase the font size, add a title, change the background colour and set display units (billions or millions)

Page 1: Overview

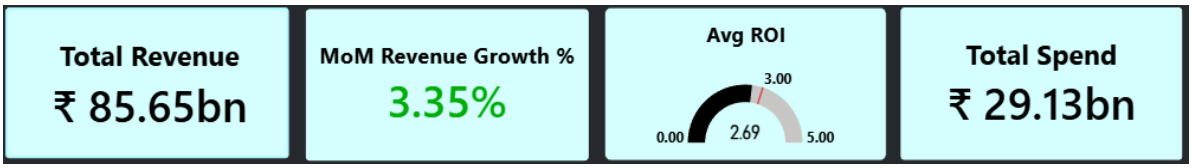


Figure 4: Overview page KPI cards

Interpretation: Campaigns earned ₹85.65bn in revenue on ₹29.13bn of spend, which is about ₹2.94 of revenue for every ₹1 spent. Average ROI is 2.69 against a target of 3.00, and revenue growth is shown month over month.

Revenue and MoM Growth *(Line and clustered column chart)*

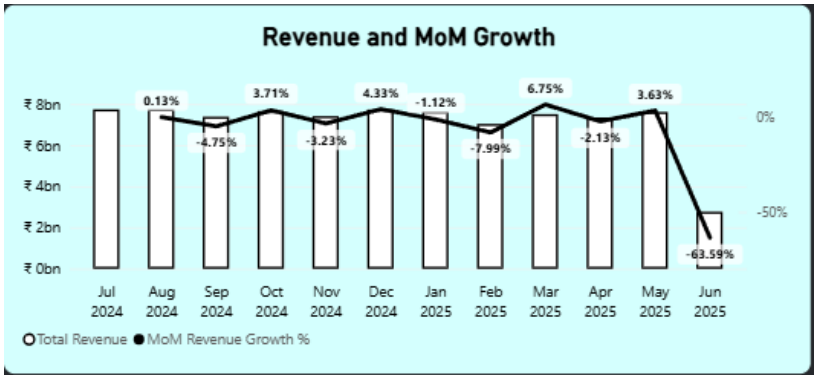


Figure 5: Revenue and MoM Growth

Interpretation: Monthly revenue stays steady at roughly ₹7–8bn, and growth moves between about −8% and +7% from month to month. The sharp fall in June 2025 is because the data ends on 24 June, so that month is incomplete.

Revenue by Campaign Type *(Ribbon chart)*

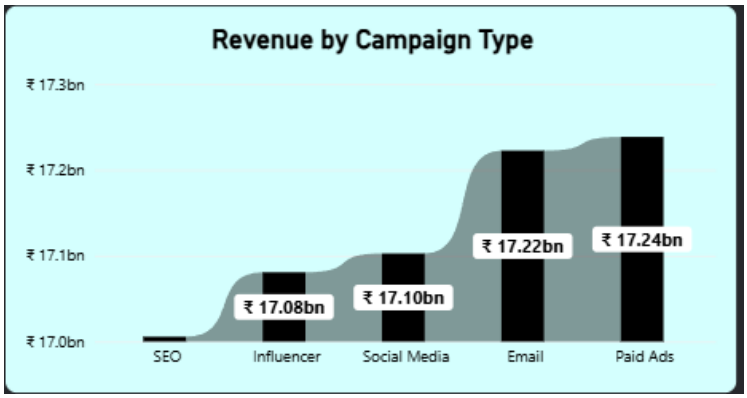


Figure 6: Revenue by Campaign Type

Interpretation: Paid Ads is the highest revenue campaign type at ₹17.24bn, followed by Email (₹17.22bn). The axis starts at ₹17.0bn, so the gaps look larger than they are; the five types are within about 1.5% of each other.

Revenue Share by Brand *(Donut chart)*

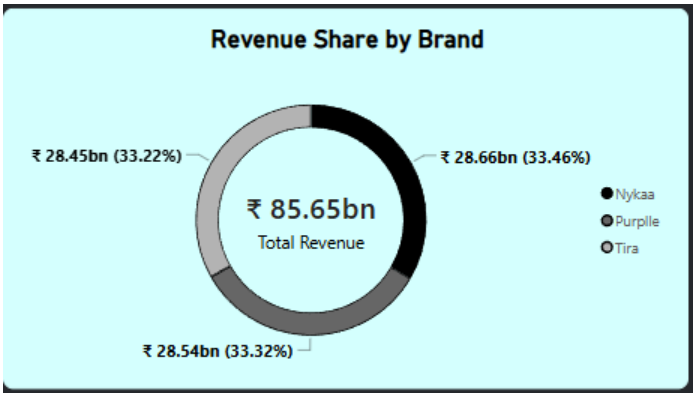


Figure 7: Revenue Share by Brand

Interpretation: Revenue is split almost evenly: Nykaa 33.46%, Purplle 33.32% and Tira 33.22%, so no brand dominates.

Revenue by City *(Map)*

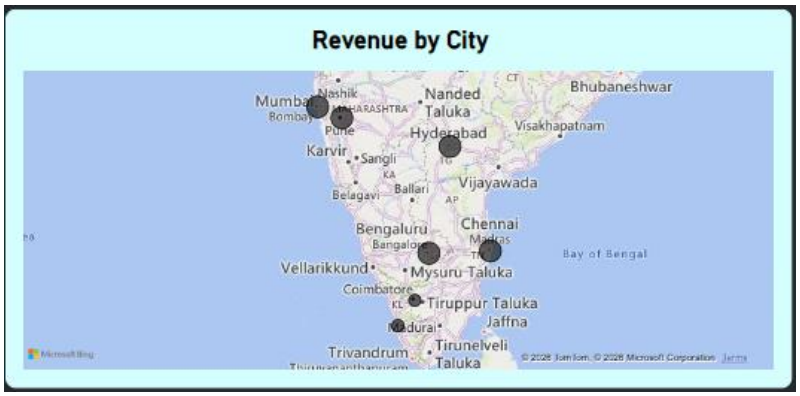


Figure 8: Revenue by City

Interpretation: Bubble size shows revenue by city. Revenue is spread across many cities, and the map shows which ones contribute most. This map is also the starting point for the drill-through.

Page 2: Performance



Figure 9: Performance page KPI cards

Interpretation: About 8.5 of every 100 impressions become clicks (CTR 8.50%). Each ₹1 of spend returns ₹2.94 (ROAS), the average engagement score is 13.77, and winning one conversion costs about ₹169.83.

Cost per Conversion by Channel *(Bar chart with average line)*

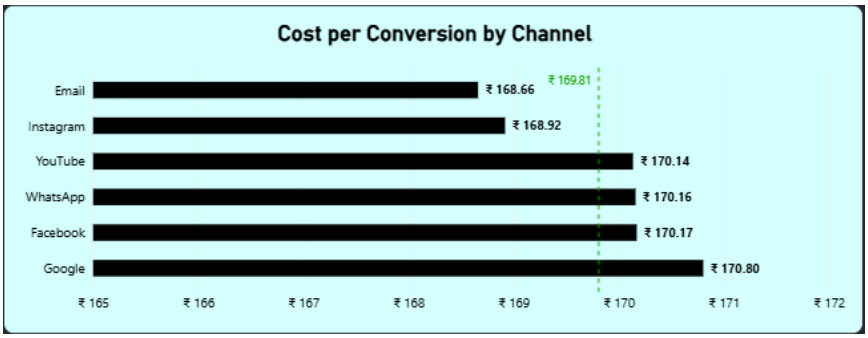


Figure 10: Cost per Conversion by Channel

Interpretation: Email (₹168.66) and Instagram (₹168.92) are the cheapest channels per conversion, and Google is the most expensive (₹170.80). The axis starts at ₹165, so the differences are small (about ₹2).

Avg ROI by Audience and Brand *(Matrix with conditional formatting)*

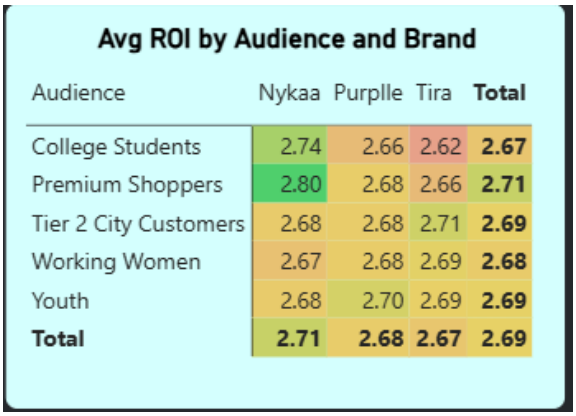


Figure 11: Avg ROI by Audience and Brand

Interpretation: Nykaa with Premium Shoppers has the highest ROI (2.80, dark green), and Tira with College Students has the lowest (2.62, red). Most other cells sit close to 2.7.

Spend vs Revenue by Campaign Type *(Scatter chart)*

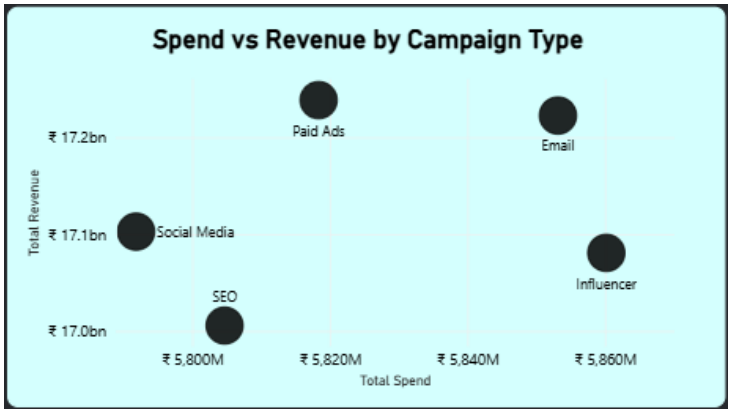


Figure 12: Spend vs Revenue by Campaign Type

Interpretation: The five campaign types sit in a tight cluster. Paid Ads earns the most revenue for relatively low spend, while Influencer has the highest spend with lower revenue.

Avg ROI by Campaign Type and Brand *(Stacked area chart)*

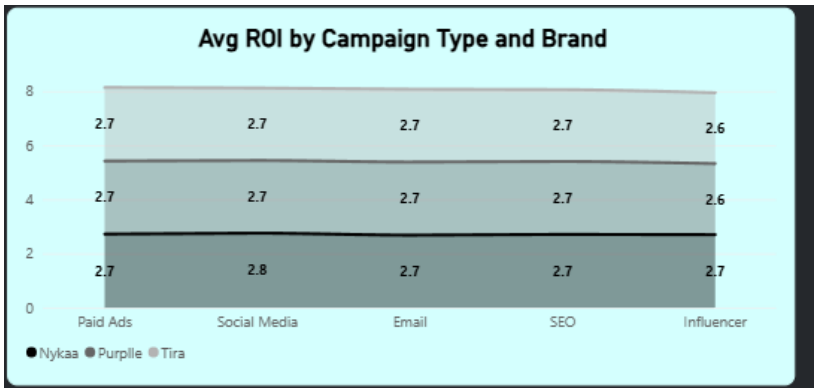


Figure 13: Avg ROI by Campaign Type and Brand

Interpretation: Average ROI is close to 2.7 for every brand across every campaign type, with Social Media for Nykaa slightly higher (2.8) and Influencer for Purplle and Tira slightly lower (2.6).

Page 3: Funnel



Figure 14: Funnel page KPI cards

Interpretation: 9.18bn impressions led to 780M clicks, 312M leads and 172M conversions, which shows how many people remain at each stage of the funnel.

Conversion Funnel *(Funnel chart)*

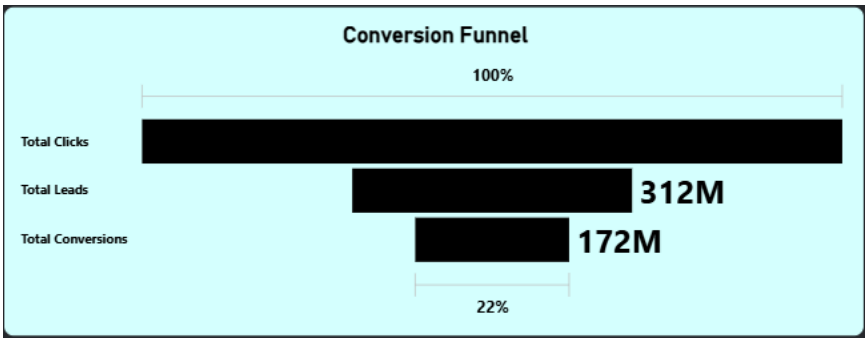


Figure 15: Conversion Funnel

Interpretation: Of 780M clicks, 312M become leads (about 40%), and 172M of those leads convert (about 55%). Overall, 22% of clicks end in a conversion.

Conversion Mix by Audience *(Treemap)*

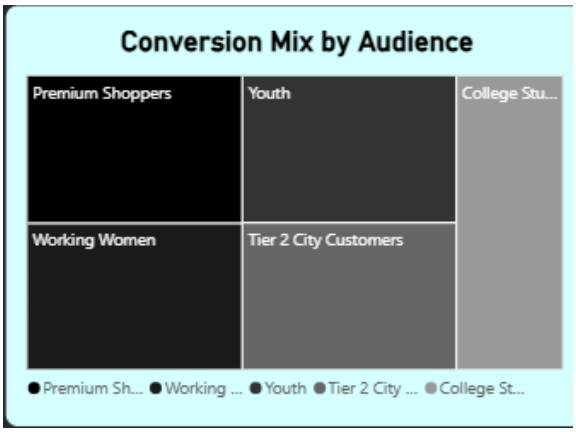


Figure 16: Conversion Mix by Audience

Interpretation: The five audience segments have tiles of similar size, so conversions are spread fairly evenly across audiences rather than coming from one segment.

Stage Rates by Campaign Type *(Table)*

Stage Rates by Campaign Type

Campaign_Type	CTR %	Lead Rate %	Lead to Conversion %	Click to Conversion %
Email	8.50%	40.01%	55.07%	22.03%
Paid Ads	8.50%	39.93%	55.05%	21.98%
SEO	8.47%	39.97%	54.98%	21.98%
Social Media	8.53%	40.01%	54.90%	21.97%
Influencer	8.52%	39.93%	54.93%	21.93%
Total	**8.50%**	**39.97%**	**54.99%**	**21.98%**

Figure 17: Stage Rates by Campaign Type

Interpretation: Email has the best click-to-conversion rate (22.03%) and Influencer the lowest (21.93%). The stage rates are nearly identical across all campaign types, so the drop-off is not specific to one type.

Conversions Over Time by Brand *(100% stacked column chart)*

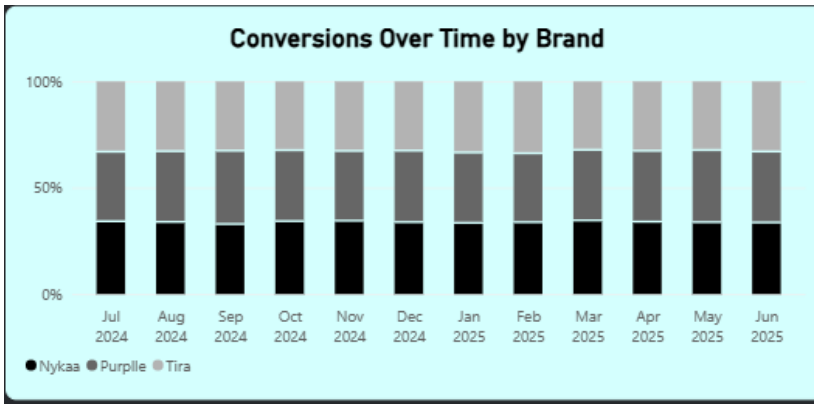


Figure 18: Conversions Over Time by Brand

Interpretation: Each brand contributes about one-third of conversions in every month, and the shares stay stable from July 2024 to June 2025.

Step 6: Add Filters & Slicers

Steps:

Channel, Campaign Type, City and Brand slicers

- Select the **Slicer** visual
- Drag the field (Channel, Campaign_Type, City or Brand) into the slicer
- Format as a **Dropdown** to save space

Date slicer

- Add a slicer and drag the Date field from Dim_Date
- Change it to **Between (range)** so a start and end date can be chosen
- Use **View → Sync slicers** so selections carry across all three pages

Now all charts update dynamically based on the slicer selections.

6. Screenshots of Dashboards

The final dashboard has KPI cards at the top, charts in the middle (each with a title and a consistent colour scheme), and slicers on the left side.



Figure 19: Screenshot 1: Overview dashboard

Shows revenue, spend, ROI and growth KPIs, the monthly trend, campaign type and brand comparisons, and the city map.

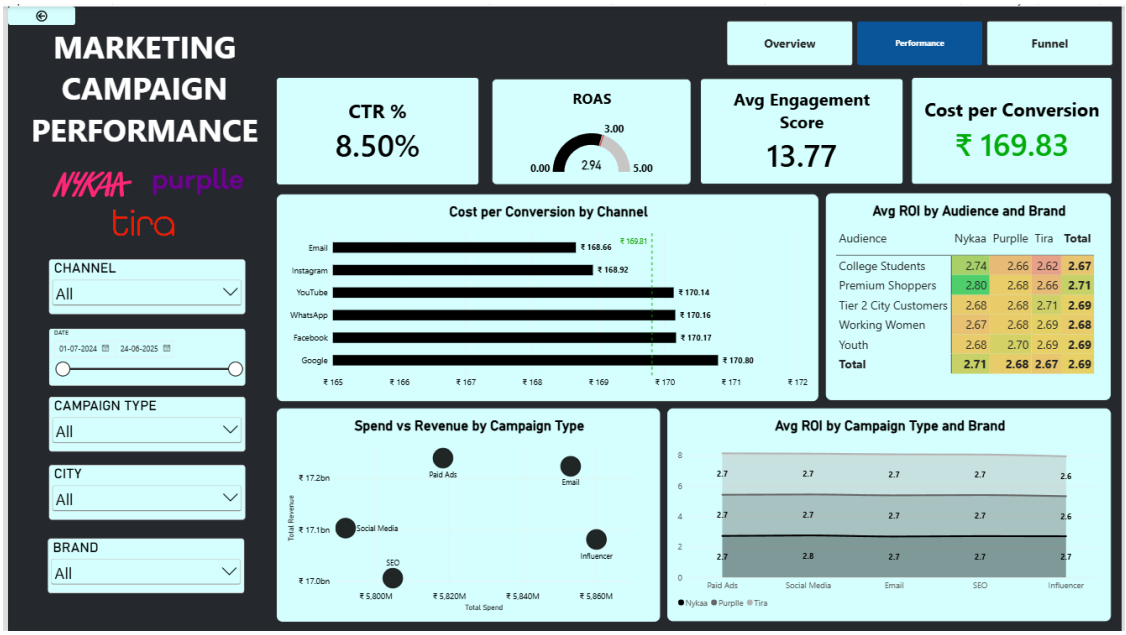


Figure 20: Screenshot 2: Performance dashboard

Shows efficiency KPIs, channel cost comparison, the ROI heatmap, spend versus revenue, and ROI by brand.

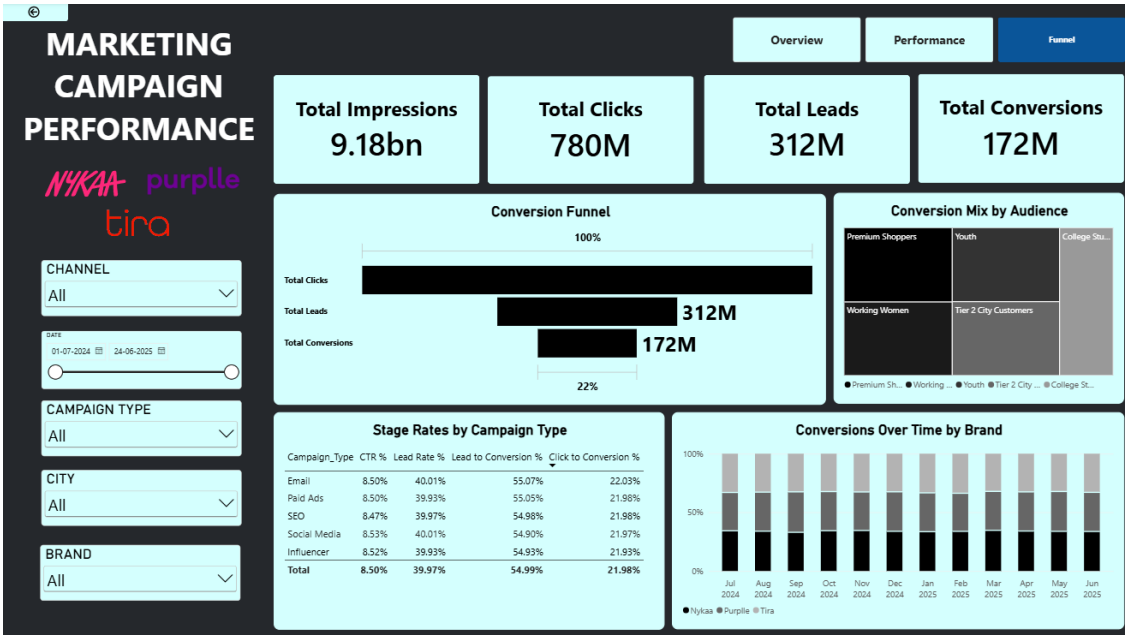


Figure 21: Screenshot 3: Funnel dashboard

Shows funnel stage KPIs, the conversion funnel, conversion mix by audience, stage rates and conversion trend by brand.

7. Case Study Analysis

Case Study 1: High Click Volume but Low Lead Conversion

Problem:

Campaigns generate 780M clicks, but only 312M become leads. It is unclear where the biggest loss in the funnel occurs.

Analysis:

- The funnel chart shows that only about 40% of clicks become leads, while about 55% of leads convert
- Overall, 22% of clicks end in a conversion
- The Stage Rates table shows the lead rate is about 40% for every campaign type, so the loss happens at the click-to-lead step across the board

Solution:

- Improve landing pages and lead forms to move more clicks to leads
- Use clearer calls to action and shorter sign-up steps
- Retarget visitors who clicked but did not become leads

Case Study 2: Consistent Performance Across Brands and Campaign Types

Problem:

Management wants to know which brand or campaign type should receive more budget.

Analysis:

- Revenue is split almost equally across brands (33.2% to 33.5%)
- Paid Ads has the highest revenue (₹17.24bn) and sits at the top of the scatter plot for revenue against spend
- Average ROI by campaign type ranges only from about 2.65 to 2.72, and the heatmap shows Nykaa with Premium Shoppers as the best combination (2.80)
- Influencer has the highest spend but lower revenue and ROI

Solution:

- Move part of the budget towards Paid Ads and the Premium Shoppers segment, and test the results
- Review Influencer spend, since it returns slightly less than other types
- Because differences are small, focus on improving creative and targeting rather than only reallocating budget

Case Study 3: Channel Cost Efficiency

Problem:

Identify which channels win conversions at the lowest cost.

Analysis:

- Email (₹168.66) and Instagram (₹168.92) are the cheapest channels per conversion
- YouTube, WhatsApp and Facebook are close to the average of about ₹170
- Google is the most expensive at ₹170.80

Solution:

- Give more budget to Email and Instagram where cost per conversion is lowest
- Review Google bidding and targeting to bring its cost down
- Keep tracking the gap, since the difference is small (about ₹2)

8. Conclusion

This project shows how Power BI can turn raw campaign data into clear business insight. Three brand files were cleaned and combined in Power Query, modelled as a star schema with a bridge table and a date table, and analysed with 19 DAX measures. The three-page dashboard presents the results through KPI cards, varied charts, slicers, navigation and drill-through.

The analysis found that campaigns earned ₹85.65bn on ₹29.13bn of spend (ROAS 2.94), that performance is very even across brands, campaign types and channels, and that the biggest funnel loss is between clicks and leads. Data for June 2025 runs only to 24 June, so growth for that month should be read with care.

The dashboard enables:

- Better decision-making on budget and channel mix
- Continuous tracking of campaign performance
- Identification of improvement opportunities in the conversion funnel

Overall, this project highlights the importance of data analytics in modern marketing.